

Predicting Spotify Track Popularity through Bayesian Reasoning

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Abstract

This mini-project aims to apply Bayesian networks to understand what influences and determines song popularity. We define a small, discrete Bayesian network built on seven variables derived from the Spotify Tracks dataset: genre, tempo, danceability, energy, valence, explicit content, and popularity. Parameters are fit with Maximum Likelihood Estimation, and we use exact inference – checked against structure-learning algorithms and approximate inference methods – to answer several questions about what actually drives a track's popularity. We find that energy turns out to be a much stronger driver of popularity than mood (valence), and that genre alone can shift the odds of a track being popular quite a bit.

Introduction

Domain

Streaming platforms like Spotify give each track a set of audio features computed directly from the signal (danceability, energy, valence, tempo, etc.), along with a popularity score. This project looks at the probabilistic relationships between a track's genre, tempo, audio characteristics and its popularity, using the Spotify Tracks Dataset (Pandya 2023).

Aim

The goal of this project is to build a Bayesian Network that connects a track's genre, tempo and audio characteristics to its popularity, and then use it to answer a few questions:

- Does genre by itself change the odds of a track being popular?
- Does mood (valence) matter more than energy for popularity?
- Are audio features connected to each other beyond just genre (e.g. does tempo drive energy, which then drives mood)?
- How much can predicted popularity swing if we combine good vs. bad evidence across every feature at once?
- Would an automated structure-learning algorithm find a similar network, or something different?
- How does approximate (sampling-based) inference compare to exact inference, in accuracy and speed?

Method

We model the problem with a **Discrete Bayesian Network**: a directed acyclic graph (DAG) of categorical random variables, each conditioned only on its parents. We used pgmpy to define this DAG by hand and fit its Conditional Probability Tables (CPTs) via Maximum Likelihood Estimation, then used Variable Elimination for exact inference. To check these choices, we also ran two automated structure-learning algorithms (Hill-Climb Search, PC) on the same data, and compared exact inference against two approximate sampling methods (Likelihood Weighting, Rejection Sampling).

Results

Energy is the clearest single driver of popularity, well ahead of mood or genre alone. Neither automated structure-learning algorithm improved on our hand-built network, and exact inference was both faster and more accurate than the approximate methods we tried.

Model

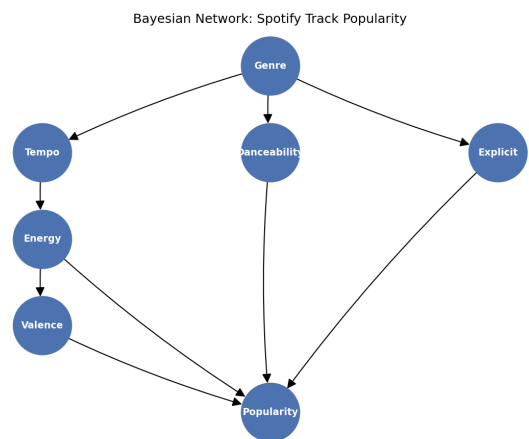


Figure 1: The Bayesian Network relating genre and audio features to track popularity.

The dataset was restricted to 6000 tracks evenly split across six genres (pop, classical, jazz, metal, edm, hip-hop),

to keep the model small and the CPTs well estimated. Continuous features (danceability, energy, valence, tempo) and popularity were discretized into three bins each using fixed range cuts, since popularity is skewed towards zero and quantile-based cuts produced degenerate bins. The seven nodes are:

- **Genre:** one of the six chosen genres.
- **Tempo:** slow/medium/fast (bpm-based).
- **Danceability, Energy:** low/med/high.
- **Valence:** sad/neutral/happy (musical mood).
- **Explicit:** boolean flag for explicit content.
- **Popularity:** low/med/high (target node).

Treating Genre as the sole cause of every audio feature would assume the audio features are mutually independent once Genre is known – not realistic: slow tracks are low-energy 47% of the time vs. 19% for fast tracks, and low-energy tracks are “sad” 51% of the time vs. 13% “happy”. We therefore built in **Genre** → **Tempo** → **Energy** → **Valence**, alongside Genre’s direct influence on Danceability and Explicit, with those four features feeding into Popularity. Popularity’s CPT stays manageable (47/54 parent combinations present, minimum count 1); the two added CPTs (Energy|Tempo, Valence|Energy) are fully populated (9/9 each, minimum count >175).

Analysis

Experimental setup

We ran exact-inference queries via Variable Elimination: (1) $P(\text{Popularity} \mid \text{Genre} = g)$ across all six genres; (2) $P(\text{Popularity} = \text{high} \mid \text{Valence} = v)$ vs. $P(\text{Popularity} = \text{high} \mid \text{Energy} = e)$; (3) a best-case vs. worst-case combined scenario; (4) $P(\text{Popularity} = \text{high} \mid \text{Tempo} = t)$; and (5) a diagnostic query, $P(\text{Genre} \mid \text{Popularity})$. We also tested for an intercausal (“explaining-away”) effect between Energy and Explicit, ran Hill-Climb Search and PC on our data to compare against our manual structure, and compared exact inference against Likelihood Weighting and Rejection Sampling at sample sizes from 100 to 50,000.

Results

Genre and combined evidence. Classical comes out weakest for popularity (50.6% low, 17.8% high), while hip-hop, edm and metal do best (24–28% high) – part of that gap comes from genre’s effect being mediated through the Tempo→Energy→Valence chain rather than acting on Popularity directly. Stacking every favorable feature together pushes this further: best-case reaches 21.7% high, against just 2.5% for worst-case.

Valence, energy, tempo. Of the two, energy is what matters: its effect on popularity is clear and monotonic, climbing from 12.6% to 24.8% to 27.9% as energy rises. Valence barely moves the needle by comparison (25.3/24.7/18.3%, not even monotonic). Tempo shows a similar pattern to energy (18.0% up to 24.7% and 24.3%), mostly because it drives energy rather than acting on popularity directly – exactly what our structure assumes.

Diagnostic and intercausal reasoning. Running the query in reverse, classical’s share drops from 18.4% (given low popularity) to 12.7% (given high popularity). We also found a genuine explaining-away effect between two of Popularity’s parents: $P(\text{Energy} = \text{high} \mid \text{Popularity} = \text{high}) = 54.3\%$ drops to 46.7% once we learn the track is also explicit – explicit content is itself a plausible reason for popularity, so less credit is needed from energy once we know it.

Structure learning. Letting an algorithm learn the structure didn’t beat building it by hand: Hill-Climb introduced backwards edges (e.g. Energy→Genre), and PC returned a near-complete, largely undirected graph (19 of 21 possible edges) – both harder to justify than our own structure.

Exact vs. approximate inference. Variable Elimination answered every query in milliseconds and matched or beat Likelihood Weighting and Rejection Sampling in accuracy at every sample size we tried, up to 50,000 samples (seconds, not milliseconds). Approximate inference isn’t worth the trade-off at this network’s scale.

Conditional independence. Popularity’s four parents imply Genre is d-separated from Popularity given those features, and the Tempo→Energy→Valence chain implies Genre is also d-separated from Valence given Energy alone; we confirmed both via exact reachability analysis – properties of the DAG we chose, not facts the data proves alone. The same logic shows in each node’s Markov blanket: Popularity’s blanket is exactly its four parents (no children), while Energy’s spans 5 of the other 6 nodes, sitting centrally with a parent, two children, and co-parents of those children.

Conclusion

Energy, not genre or mood, is the main driver of predicted popularity, and letting audio features depend on each other (rather than assuming independence given genre) changes which pathways carry that influence. Checking the design against alternatives – automated structure learning and approximate inference – didn’t turn up anything better, which supports the choices made here. Limitations: continuous features were discretized into only three bins each; the dataset was restricted to six genres out of 114 for tractability; Maximum Likelihood Estimation falls back to a uniform distribution for the few parent combinations missing from our sample; and the source dataset samples exactly 1000 tracks per genre, giving Popularity’s Genre prior an artificial uniformity that doesn’t reflect real-world catalog proportions.

Links to external resources

- Dataset: <https://www.kaggle.com/datasets/maharshipandya/spotify-tracks-dataset>
- Code repository: <https://github.com/MattiaSemproli/Spotify-Bayesian-Network>

References

Pandya, M. 2023. Spotify tracks dataset. Kaggle.